

AnyCam 2.6.5 Audit Report

Release audit - 02 October 2026 - Enhanced View changes - tag 2.6.5

2.6.5 restores motion detection on the Lorex channels and makes it work with no viewer (build plan B1, C2). It changes Enhanced View in four ways CrystalHeeler asked for on 2026-09-29: slow-starting cameras play live, landscape on a phone fills the screen, the Resolution, Frame Rate and Auto menus are gone, and a loading message replaces the blank or gray screen. It closes B14, and the zip folder no longer carries the version (F8). camera_discovery.py against 2.6.4: 386 lines added, 328 removed. A companion release, Oak-D camera add-on 2.4.2, fixes the slow-starting stream at its source.

Investigation before code

Motion detection stopped working (B1)

CrystalHeeler, 2026-09-29, on 2.6.4: ch7 of the Lorex DVR was armed and no clip was written. His log shows every Lorex card polling HTTP snapshots ("http frame" lines). Motion detection was called only on the ffmpeg path (`_motion_on_frame` did not exist; the check sat inline in `snap_loop`). The HTTP loop has had no motion check in any version, checked back to the archived 2.5.0.

CrystalHeeler said it had worked on 2026-09-28, on the Lorex cameras only. Log1 (that date) settles it: ch7's card was on HTTP polling, and ffmpeg frames appear only while ch7 was open in the classic Enhanced View. Motion worked there, and only there. 2.6.4 made live view the default, so the classic view stopped running and so did motion detection: a regression from our change, as rule 8 predicts. The same mechanism explains B1: leaving the view ended the only loop that could stop a recording.

I first told CrystalHeeler this was not a regression, reasoning from the HTTP loop's history alone. Log1 corrected it. CrystalHeeler also recalled an earlier discussion, and found it: during the 2.6.3 build (2026-09-29 03:04) I noted that the snapshot loop stops 30 s after the last poll, so armed cameras would lose motion recording, and fixed it for live view only: `_focus_set_go2rtc` starts the thumbnail loop when motion is armed, and the card poller keeps it alive while the page is open. That fix assumed the thumbnail loop checks motion, which is true only on the ffmpeg path; on the Lorex channels it kept alive an HTTP loop that never checked. My first search missed it because it read only my replies, not my reasoning blocks. 2.6.5 keeps the 2.6.3 behaviour and extends it.

The Oak-D H.264 camera (B13)

CrystalHeeler saw this camera drop to poor quality on the phone, with the Connecting to RTSP stream message. The 2026-09-29 log from test system B (2.6.3) gave the cause. Every new connection to this stream was timed from start to first frame:

Result	Connections	Time to first frame
Produced a frame	12	19 s shortest, 23 s median, 28 s longest
Gave up first	13	at 12 s (live view) or 30 s (classic)

The phone session at 12:19: live view was not tried, because a failure at 07:42 was remembered until the page reloaded, and the HA app kept the page open all day. The classic view then stopped the card's working stream (frame 38,450) and opened a new connection, which timed out after 30 s with no frames. The poor quality was the card thumbnail left in the buffer.

go2rtc logged no errors. The source of the stream is the Oak-D camera add-on (version 2.4.1, on the build PC). Its ffmpeg command runs `libx264` with no `-g` option:

```
-c:v libx264 -preset ultrafast -tune zerolatency -b:v 1000k
```

x264's default is one keyframe every 250 frames. A player cannot draw until the first keyframe, so a long interval gives every viewer a long start. The fix belongs in that add-on and needs CrystalHeeler's order.

No changelog in Home Assistant

CrystalHeeler's folder listing showed CHANGELOG.md present in /addons/camera_discovery-2.6.4/. Supervisor source gives the cause: on a store reload it refreshes each add-on's file checks before it reads the new folder location (supervisor/store/__init__.py, reload()), so the check runs against the deleted 2.6.3 folder and caches no changelog. The versioned folder name (CLAUDE.md rule 3) triggers it. No AnyCam code is involved; the workaround and the choice of lasting fix are in the changelog and build plan F8.

Landscape: what the page can do

Home Assistant's app panel (home-assistant/frontend, src/panels/app/ha-panel-app.ts) accepts a subscribe-properties message with kioskMode from the add-on page and hides its title bar; the unsubscribe message restores it. Present since the January 2026 app panel. The panel's iframe has no allow="fullscreen", so the browser Fullscreen API is not available.

Changes

Change	Detail
Live view waits 30 s	Up from 12 s, CrystalHeeler's choice. Covers the 28 s longest start measured.
Live view retries	Only a codec or mode error is remembered for the page's life (Firefox and H.265, C10). A timeout, a dropped connection or an error that names no mode is not.
Landscape full screen	Touch screens only: (orientation: landscape) and (pointer: coarse). Hides the bottom bar and sends the kiosk request; portrait or close sends the release. The message targets the page's own origin.
Menus removed	Resolution, Frame Rate and Auto, their JS (focusPickRes, focusPickFps, focusResetAuto, _loadFocusProfiles, _applyFocusTier, _go2rtcSwitchProfile) and CSS. Classic button and stream information kept. "Decoded on this device" text removed.
Old manual tier cleared	The server kept a manual tier between sessions and the page can no longer clear it. handle_focus_set now drops manual_override on entry. Learned adaptive locks stay.
Loading message	"Loading feed, please wait..." until the first frame, both engines. Live player poster set to a 1x1 transparent GIF (the Android WebView gray play icon). Classic hides the thumbnail until the server's new X-Focus-Frames header is above 0.
Connecting status	Before the first frame it shows as the loading message's second line instead of the warning box.
Cards (B14)	"Loading feed, please wait..." while starting; "Stream unavailable" only after 90 s of errors, up from 3 errors (about 16 s).

Motion detection

Change	Detail
Both paths	_motion_on_frame is called from the ffmpeg loop and the HTTP snapshot loop.

Change	Detail
No viewer needed	All four idle stops skip an armed camera.
Keeper	_motion_keeper, every 10 s: restarts a dead loop for an armed camera (not during the classic view, which runs its own), stops a recording after cooldown plus padding whether or not frames arrive (B1), and clears a recording whose ffmpeg exited.
Persistence	/data/motion.json holds the armed list; restored at start for cameras that still exist.
Record button	Field test 2026-09-30: the server restored ch7 after both restarts, but the card said Record, and a click at 07:20:45 disarmed it. The page never read the server's state, and the button flipped it blindly. Now: GET /api/motion for all cameras at load and every 3 s; the button sends the state asked for; no body keeps the old flip.
Recording stderr	Drained, so a full 64 KB pipe cannot block ffmpeg mid-recording.
False triggers	The comparison frame resets on every ffmpeg launch and HTTP loop start: a resolution change read as motion.
Throttle	An armed camera is not paused while another camera is focused in live view (no Pi decode). The classic view still pauses others.
Cost	Each armed Lorex channel polls a snapshot every 1.8 s (measured) at all times.

Packaging (F8, CrystalHeeler's order)

The zip keeps its versioned name; the folder inside is always local_camera_discovery. CLAUDE.md rule 3 is changed with the reason written in. The first update into the new folder can still show no changelog once; the changelog gives the one-time touch command.

Oak-D camera add-on 2.4.2 (separate project)

On CrystalHeeler's order: -g set to the FPS option, one keyframe per second. Committed and tagged v2.4.2 in its own repository; packaged as oak_camera_app-2.4.2.zip with folder oak_camera_app, the name on the Pi. Syntax-checked only; no Oak-D tests exist.

Held back: fix 1c

Proposed and approved: keep the card's stream until the classic view's new stream delivers. Not built. It needs two connections to one camera at once, through per-camera state (_SNAP) that holds one process and one task. Rate-limited cameras such as the Microseven refuse a second connection. Build plan B15, to discuss.

Tests run

Suite	Result	Scope
Release gate	PASS	All five gates; 24 function contracts; go2rtc API host check.

Suite	Result	Scope
Python integration	PASS	120 of 120, up from 81. Section G: manual tier dropped on entry, learned lock kept, X-Focus-Frames 0 then 1, 8 page checks. Section H (motion): detection and stop, keeper stops a recording with no frames, restarts a loop, clears a dead recording, respects the classic view; toggle saves, restart restores; and the real HTTP snapshot loop against a local server keeps polling with no viewer, runs every frame through detection, starts a recording on a size jump, and stops once disarmed and idle. H6, through a real server: set-state is idempotent, the all-cameras map, the no-body flip.
JavaScript behaviour	PASS	75 of 75, up from 46. Real vendored player class. New: blank poster, 30 s watchdog, a first frame at 23 s plays, what is and is not remembered, landscape kiosk messages and origin, and the classic loading gate driven through the real openFocus, poll loop and closeFocus. Section R: cards start with the server's motion state, the button asks for the opposite of what it shows, and changes from another device redraw the card.
Page build	PASS	node --check on the full page script.
Visual check	PASS	Built page in a browser at 390x760 and 760x360 touch emulation: loading message centred; landscape picture fills the height, keeps 16:9, black side bars, X visible, bar hidden. The kiosk request needs a Home Assistant parent frame: covered by the JS test only.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*. CLAUDE.md rule 1 requires this section in every audit. It was present through 2.3.2 and dropped during the 2.4.x cycle. It is restored and stays.

Scope note. That document contains Parts 1 to 5 and a Sources Summary. It has no Part 6. CLAUDE.md describes a "Part 6 - Project Engineering" covering security, dependency management and version control, so the working agreement and the document disagree. Rows below marked *outside Parts 1-5* were checked against ordinary engineering practice, not against a written AnyCam standard.

Standard	Check applied	Result	Note
1.1 Truncated function	AST plus duplicate-def scan	PASS	161 top-level definitions, no duplicates.
1.2 Syntax errors	compile() to bytecode	PASS	Gate 1.
1.4 Async/await	focus_frame_base; keeper; file I/O	PASS	The cancelled task cannot add to frame_count again. The keeper is kept in _MOTION_TASK and cancelled first at shutdown. The toggle writes motion.json through asyncio.to_thread.
1.9 Exceptions	Keeper and persistence	PASS	One camera's error cannot stop the keeper; load catches OSError and ValueError only.
1.5 Dead code	Removed page controls	NOTED	Seven JS functions removed; the inline motion block replaced by one helper. The server endpoints /snap/focus/tier and /snap/focus/profiles now have no caller; kept this release, build plan E8.
1.5 Line length	Measured every added line	NOTED	27 of 386 added lines exceed 79 characters: 14 Python (80 to 99), 13 JS and CSS inside the page strings, matching the file's existing style.
Part 2 JavaScript	No await in forEach; listener fallbacks	PASS	matchMedia change listener with addListener fallback; postMessage in try/catch.
Part 3 HTML	Markup	PASS	Loading element added; focus image alt emptied because the loading text now describes the state.
Part 4 CSS	100dvh with 100vh fallback	PASS	Landscape height uses both, as Part 4 asks.
5.1 f-string braces	Rendered CSS checked	PASS	Integration test asserts single braces in the built page.

Standard	Check applied	Result	Note
Outside Parts 1-5: security	Cross-frame message	PASS	postMessage targets window.location.origin, not *. The page sends only the two documented HA message types.

Known issues carried forward

- **The Oak-D camera needs about 20 s to start** until Oak-D 2.4.2 is installed (B13).
- **Recording length** is one file per motion event (C12).
- **Light changes trigger recordings** (C13): 06:21 on 2026-09-30, a 9 s clip at the night-to-day switch. JPEG-size method unchanged.
- **Opening the classic view stops the card stream** (B15).
- Unchanged: Firefox and H.265 (C10); Skip Non-Reference Frames (B3); hardware decode falls back to software (B11); a motion recording may not stop (B1).

Not verified

- **Kiosk mode on the real HA app.** The message format is from HA's source and the JS test; the phone test is A12.
- **The image was not built here.** The Dockerfile is unchanged from 2.6.4.
- **The changelog workaround** is inferred from Supervisor source.
- **Motion on real cameras.** The loop and keeper ran against a local test server; a real clip on a Lorex channel is the A12 check.

Acceptance

- Arm a Lorex channel, close the page, make motion: a clip appears in /media/anycam and ends about 10 s after the motion stops. The channel is still armed after a restart.
- HA app, Enhanced View, landscape: picture fills the screen, AnyCam title bar and bottom bar hidden, red X visible. Portrait restores both.
- No Resolution, Frame Rate or Auto controls; Classic button present.
- "Loading feed, please wait..." instead of the gray play icon.
- The Oak-D camera plays live after about 20 s in Chrome, including a second open in the same session.
- Cards show "Loading feed, please wait..." while starting.

Files changed

File	Change
camera_discovery.py	Enhanced View JS, markup and CSS; card placeholder and error timing; handle_focus_set clears manual tiers and records a frame base; X-Focus-Frames header; motion helpers, keeper, persistence, idle and throttle changes. Version 2.6.5.
config.yaml	Version 2.6.5.
CHANGELOG.md	2.6.5 entry.
CLAUDE.md	Rule 3: fixed folder name local_camera_discovery, with the reason.
docs/BUILD_PLAN.md, .html	A12, B13, C1, C3 updated; B1, B7, B14, C2, F8 done; B15, C13, E8, F9 added.